

Midterm Exam: CS 215

Attempt all six questions. You have a time of 120 minutes for this exam. Clearly mark out rough work. No calculators or phones are allowed (or required :-)). You may use results/theorems we have stated or derived in class, unless explicitly stated otherwise. Avoid writing lengthy answers.

Useful Information

1. The empirical mean of n independent and identically distributed random variables is approximately Gaussian distributed. The approximation accuracy is better when n is larger. If the random variables are Gaussian, the empirical mean is exactly Gaussian distributed.
2. For a non-negative random variable X , we have $P(X \geq a) \leq E(X)/a$ where $a > 0$.
3. For a random variable X with mean μ and variance σ^2 , we have $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$.
4. Gaussian PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$, MGF $\phi_X(t) = e^{\mu t + \sigma^2 t^2/2}$
5. Poisson PMF: $P(X = i) = \frac{e^{-\lambda}\lambda^i}{i!}$
6. Taylor series expansion of $f(x)$ about x_0 is given as $f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2 f''(x_0)}{2!} + \dots + \frac{(x - x_0)^n f^{(n)}(x_0)}{n!}$

1. Consider random variables $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$. Let Y be a random variable that takes on the value of X_1 with probability p and the value of X_2 with probability $1 - p$ where $0 \leq p \leq 1$. Write down an expression for the PDF of Y in terms of the PDFs of X_1 and X_2 . If you had access to a program to draw a sample value from $\mathcal{N}(0, 1)$, and a program to draw a sample value from $\text{Uniform}(0, 1)$, then state a procedure to draw a sample from the distribution of Y . [17 points]

2. If $X \sim \text{Uniform}(a, b)$ where $0 < a < b$, derive the mean, median, variance, PDF and CDF of $Y = \frac{1}{X}$. [17 points]

3. Let $X \sim \text{Poisson}(\lambda)$. In this question, we present a method to prove that the variance of $\sqrt{X}$ is approximately 0.25 when λ is large. (This is popularly called the variance stabilizing transform.) To this end, we define $g(X) = \sqrt{X}$. Write down the second order Taylor series expansion of $g(X)$ about λ . (For the Poisson distribution, the higher order terms can be ignored for large values for λ . This is a fact, which you are not expected to verify here.) Taking expectation on both sides, derive the expression for $E(g(X))$ and hence the expression for the variance of $g(X)$. [4+8+5=17 points]

4. Consider n sample points $\{(x_i, y_i)\}_{i=1}^n$ where for all i , y_i is the value of a random variable $Y_i \sim \mathcal{N}(\alpha + \beta x_i, \sigma^2)$, x_i is known, and α, β are unknown. Assume that the random variables $Y_1, Y_2, \dots, Y_n$ are independent. Given these sample points, we have seen in class that the least squares estimate of β and α are respectively given by $\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})Y_i}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$ and $\hat{\alpha} = \frac{1}{n} \sum_{i=1}^n Y_i - \hat{\beta}\bar{x}$ where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. Show that these estimates are unbiased. Derive an expression for their variance in terms of $n, \bar{x}, \sigma^2, \{x_i\}_{i=1}^n$. Note that the values $\{x_i\}_{i=1}^n$ are treated as constants. [5+5+4+3=15 points]

5. A storage device contains the annual income of a group G of n families in a country. A computer program has read through these records, and has computed and stored in memory the value $S = \sum_{i=1}^n \sum_{j=1}^n (x_i - x_j)^2$ where x_i is the annual income of the i^{th} family. Some analysis you wish to perform requires the sample standard deviation of the annual income of the families in G . However, the storage device is incredibly slow and you do not have the option of reading any of the data again. How will you compute the standard deviation given S and n ? Derive all required formulae if necessary. [17 points]
6. Given sample values $x'_1, x'_2, \dots, x'_n$ respectively from $n > 0$ iid random variables $X_1, X_2, \dots, X_n$, each having the CDF $F_X(x)$, the so-called empirical CDF is defined as $F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(x'_i \leq x)$ where $\mathbf{1}(q)$ is the indicator function which produces 1 if the predicate q is true, and 0 otherwise. Prove that $F_n(x)$ is an unbiased estimate of $F_X(x)$ and derive its variance. Hence prove that $\lim_{n \rightarrow \infty} E([F_n(x) - F_X(x)]^2) = 0$. [6+7+4=17 points]